BATTERY CONDITION INDICATOR AND AUTOMATIC DUAL-VOLTAGE ADAPTER

IC4484B911AA FOR 24 VOLTS IC4484B911AB FOR 36/48 VOLTS

Before any adjustments, servicing, parts replacement or any other act is performed requiring physical contact with the electrical working components or wiring of this equipment, DISCONNECT THE BATTERY AND DISCHARGE CAPACITOR 1C.

FUNCTION

The function of the Battery Condition Indicator is to monitor the battery volts, to turn on a warning light if the voltage indicates a discharged battery and a fixed time later open the hydraulic pump contactor coil circuit. The 36/48 volt form of this same control will automatically change its reference to correctly monitor either a 36 volt battery or a 48 volt battery, and will energize a relay coil circuit if needed to adapt other accessories for 36/48 volt operation.

OPERATION

See page 2 for schematic diagram.

1. IC3A delays the energizing of the pump contactor coil for .15 seconds so that IC2C can distinguish between 36 and 48 volts and so that IC3B can determine if the input ever reaches nominal battery volts.

2. If the voltage exceeds 42V, ±5%, SR1 will be gated on, picking up VR to insert resistance in other coils or horn circuits, if needed for 48 volt operation on a 36/48 volt panel.

3. If the battery does not reach 95 to 100% of nominal battery volts, the output of IC3B remains zero volts, biasing Q5 off, preventing the pump contactor from closing, turning the indicator light on and locking Q5 off after a further delay of about 90 seconds.

4. If the battery volts fall below 1.7 volts per cell as set by RH4, the output of IC2A will go to zero permitting IC2B to time. The output of IC2A can be monitored with a voltmeter from #9 to negative. If the input remains below 1.7 volts continuously for about 30 seconds, as determined by R52 and C12, IC2B output switches to zero volts, gating SR2 on, latching on the indicator light and initiating the timer described in the next paragraph.

5. About ninety seconds after SR2 is gated on, as determined by R54 and C13, the output of IC3C switches to zero volts. If terminals 10 and 7 are connected, this biases Q5 off and the pump contactor drops out. The light is latched on and the pump contactor is locked out until the battery plug is pulled, as in changing the battery.

The locking out of the pump contactor as described in paragraph 5, is intended to so limit the useful work of the truck that the operator will be forced to have the battery recharged when the battery has been discharged about 80%.

CONNECTIONS

Connect Battery Condition Indicator per sheet 3.

For 36 or 48 volts, connect a jumper from terminal 5 to terminal 11.

For 36/48 volts, when a relay is needed to insert resistance in a coil or horn circuit, add a 48 volt relay, VR. Connect coil from positive to terminal 2.

For 24 volts, remove jumper from terminal 11 and connect jumper from terminal 5 to terminal 13.

For indicating lamp, connect lamp from positive to terminal 4. The lamp will be latched on if the input voltage does not reach nominal volts or if the input remains low continuously for about 30 seconds.

If you do not desire to lock out the pump contactor about 90 seconds after the indicating light is latched on, remove the jumper from terminal 7 to terminal 10.

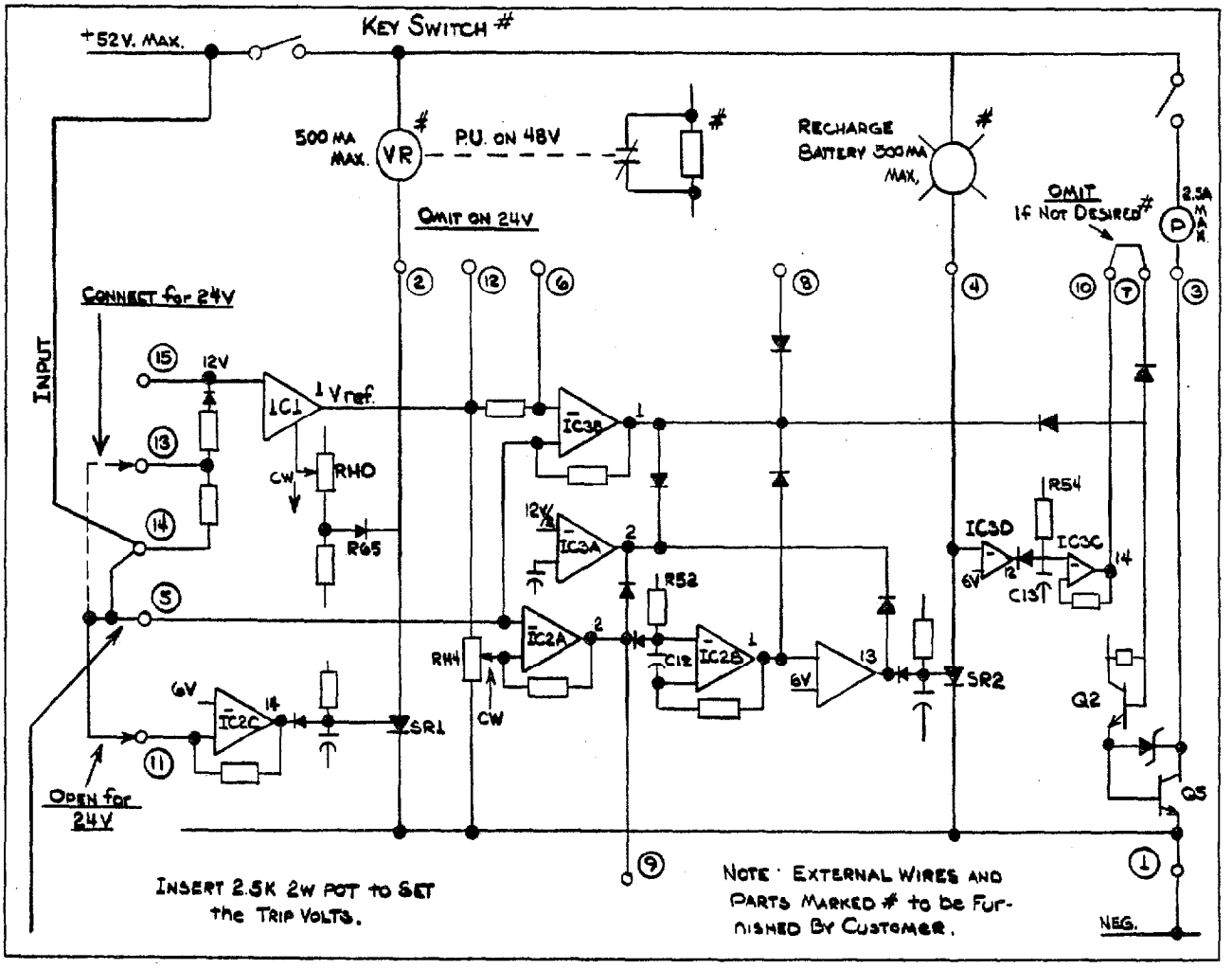
TEST

With voltmeter from terminal 12 to negative, the reference voltage Vref should be as follows:

NOMINAL BATTERY VOLTS	Vref	TRIP VOLTS
48	5.64	40.8
36	4.24	30.6
24	2.82	20.4

The reference voltage has been set at the factory for 2.82V on cards rated for 24 volts and for 4.24V on cards rated for 36/48 volts using a 36 volt input. The reference voltage may be measured with a voltmeter from terminal 12 to terminal 1. It can be varied with the trimpot RHO located in the center of the card. This requires removing the card from the box and also the use of a high accuracy digital voltmeter.

The drop out voltage can be set by inserting a 2.5K, 2W pot in the lead to terminal 5, a 1% accuracy voltmeter from terminal 5 to negative and a 15 volt or more voltmeter from terminal 9 to negative. Reduce the voltage from terminal 5 to negative to the trip volts given in the table. The voltage from terminal 9 to negative should rise to about 12 when terminal 5 is at the voltage given in the table. Adjust RH4 as needed, CW increases trip volts.



BATTERY CONDITION INDICATOR AND DUAL VOLTAGE INDICATOR

The information contained herein is intended to assist truck users and dealers in the servicing of SCR control furnished by the General Electric Company. It does not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance.

Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purpose, the matter should be referred to the truck manufacturer through his normal service channels, not directly to General Electric Company.

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